

Ian Waudby-Smith

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Academic positions

University of California, Berkeley
Miller Postdoctoral Fellow in Statistics
Host: Michael I. Jordan

Berkeley, CA
2024–2027

Education

Carnegie Mellon University
PhD, Statistics
Advisor: Aaditya Ramdas
Umesh K. Gavaskar Best Dissertation Award

Pittsburgh, PA
2019–2024

University of Waterloo
BMath, Pure Mathematics & Statistics
5-year co-op program
Dean's Honours List

Waterloo, Ontario
2013–2018

Papers

Chase Mathis and **Ian Waudby-Smith**. Confidence horizons. *Under review at the Journal of the Royal Statistical Society, Series B*.

Keegan Harris, Brian W. Lee, **Ian Waudby-Smith**, Philip Amortila, Nika Haghtalab, and Michael I. Jordan. Post-training at the edge of detectability: A game-theoretic approach to fine-tuning. *Under review at the International Conference on Learning Representations (ICLR)*, 2026.

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Gaffke's confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *Under review at the Annals of Statistics*.

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith**. Combining e-values via demi-supermartingales. *Under review at the Annals of Applied Probability*.

Ricardo J. Sandoval*, **Ian Waudby-Smith***, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925*.

Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *Under review at the Journal of the Royal Statistical Society, Series B*, 2026.

Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments. *Major revision at Biometrika*.

Ian Waudby-Smith, Ricardo J. Sandoval, and Michael I. Jordan. Universal log-optimality for general classes of e-processes and sequential hypothesis tests. *R&R at the Annals of Statistics*.

Ian Waudby-Smith, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *Major revision at Probability Theory and Related Fields*.

Ian Waudby-Smith, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform large sample anytime-valid inference and the Robbins–Siegmund distributions. *Major revision at Biometrika*.

($\alpha\beta$) Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *Bulletin of the London Mathematical Society (to appear)*, 2026.

Ian Waudby-Smith, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability (to appear)*, 2026.

Ian Waudby-Smith, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.

Ian Waudby-Smith, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/JMS Journal of Data Science* (*Invited contribution*), 1(3):1–42, 2024.

Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology* (*Discussion paper*), 86(1):1–27, 2024.

Ian Waudby-Smith, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning* (*Oral presentation*), 2023.

Akash V. Maharaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.

Ian Waudby-Smith, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting* (*Best paper award*), pages 124–139. Springer, 2021.

Ian Waudby-Smith and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems* (*Spotlight paper*), 33:20204–20214, 2020.

Ian Waudby-Smith, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.

Ian Waudby-Smith, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

($\alpha\beta$) Alphabetical ordering.

*Co-lead author.

Teaching Experience

Carnegie Mellon University

Graduate Teaching Assistant

- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

Pittsburgh, PA

2019–22

Other experience

Google Research

Student Researcher

Mentors: Jean Pouget-Abadie & Jennifer Brennan

New York, NY

Jun–Aug 2023

Microsoft Research

Research Intern

Mentor: Paul Mineiro

New York, NY & Redmond, WA

May–Aug 2022

Adobe Research

Research Intern

Mentors: David Arbour & Ritwik Sinha

San Jose, CA

Jun–Aug 2020

The Hospital for Sick Children (SickKids)

Research Student

Mentor: Eleanor Pullenayegum

Toronto, Ontario

Apr–Aug 2019

Health Data Science Lab, University of Waterloo

Research Assistant

Mentors: Joel Dubin & Joon Lee

Waterloo, Ontario

2016–18

Department of Statistics, University of Waterloo

Research Assistant

Mentor: Pengfei Li

Waterloo, Ontario

Apr–Aug 2017

Department of Pure Mathematics, University of Waterloo
Research Assistant
Mentor: Yu-Ru Liu

Waterloo, Ontario
Jan - Apr 2017

Cancer Care Ontario
Student Analyst
Mentor: Zhihui (Amy) Liu

Toronto, Ontario
Jan-Apr 2016

Service

Thesis committee member

- Tyron Lardy (PhD in Statistics, Leiden University)
- Noah Liniger (MSc in Data Science, ETH Zurich)

Reviewer

- Annals of Statistics
- Bernoulli
- Biometrika
- Electronic Journal of Probability
- Electronic Journal of Statistics
- Journal of the American Statistical Association
- Journal of the Royal Statistical Society, Series B
- Management Science
- New England Journal of Data Science
- Sankhya A.
- Symposium on Theory of Computing (STOC)

University of California, Berkeley
Volunteer

Berkeley, CA

- Miller symposium organizing committee
- Incoming Miller fellow initiative

Carnegie Mellon University
Volunteer

Pittsburgh, PA

- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

Awards

Institute of Mathematical Statistics
New Researcher Travel Award

Berkeley, CA
2026

Miller Institute for Basic Research in Science
Miller fellowship

Berkeley, CA
2024-'27

Carnegie Mellon University Department of Statistics and Data Science
Umesh K. Gavaskar Best Dissertation Award

Pittsburgh, PA
2024

Statistical Society of Canada
Probability Section Student Research Presentation Award

St. John's, NL
2024

Amazon Science
Graduate Research Fellowship

Pittsburgh, PA
2023

University of Waterloo
Waterloo Statistics Student Conference Presentation Award

Waterloo, Ontario
2022

Carnegie Mellon University Department of Statistics and Data Science
Teaching Assistant of the Year

Pittsburgh, PA
2021

Adobe Research <i>PhD Research Gift</i>	Pittsburgh, PA 2020
University of Waterloo <i>David Johnston International Experience Award</i>	Waterloo, Ontario 2018
The Natural Sciences and Engineering Research Council of Canada <i>NSERC Undergraduate Student Research Award</i>	Waterloo, Ontario 2017
University of Waterloo <i>President's Research Award</i>	Waterloo, Ontario 2016–17
University of Waterloo <i>University of Waterloo President's Scholarship of Distinction</i>	Waterloo, Ontario 2014

Presentations

Short courses.....

MBZUAI <i>A brief introduction to game-theoretic, safe, anytime-valid inference</i> Mini course consisting of 4 lectures	Abu Dhabi, UAE 2025
ERC OCEAN retreat <i>A brief introduction to game-theoretic, safe, anytime-valid inference</i> Mini course consisting of 3 lectures	Venice, Italy 2024

Department seminars.....

Department of Statistics at the London School of Economics <i>Log-optimality and multi-armed sequential hypothesis testing</i>	London, UK 2026
Department of Statistics and Actuarial Science at the University of Waterloo <i>Log-optimality and multi-armed sequential hypothesis testing</i>	Waterloo, Ontario 2026
Department of Statistics at University of California, Davis <i>Log-optimality and multi-armed sequential hypothesis testing</i>	Davis, CA 2026
Department of Statistics at Iowa State University <i>Anytime-valid inference and uniform central limit theory</i>	Ames, IA 2026
Division of Biostatistics at the University of California, Berkeley <i>Anytime-valid off-policy inference for contextual bandits</i>	Berkeley, CA 2025
Department of Statistics at Stanford University <i>Anytime-valid inference and uniform central limit theory</i>	Stanford, CA 2025
Statistics Seminar at the University of Copenhagen <i>Anytime-valid off-policy inference for contextual bandits</i>	Copenhagen, Denmark 2023

Invited talks at conferences / workshops / research groups.....

Institute for Operations Research and the Management Sciences (INFORMS) <i>Multi-armed log-optimality and sequential testing</i>	San Francisco, CA 2026
Cornell University Rising Stars in Statistics and Data Science Workshop <i>Advances in asymptotic anytime-valid inference</i>	Ithaca, NY 2026
Joint Statistical Meetings (JSM) <i>Asymptotic anytime-valid inference</i>	Boston, MA 2026
Adaptive Experimentation Workshop at University College London (Gatsby) <i>Log-optimality and multi-armed sequential hypothesis testing</i>	London, UK 2026

International Workshop on Sequential Methodologies <i>Asymptotic anytime-valid statistical inference</i> Recipient of the IMS New Researcher Travel Award	Washington, D.C. 2026
Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis) <i>On nonasymptotic confidence intervals for treatment effects in randomized experiments</i>	Stanford, CA 2026
National University of Singapore Young Mathematical Scientists Forum <i>Log-optimality of e-processes and sequential tests</i>	Singapore 2025
Stanford Data Driven Seminar (host: Stefan Wager) <i>Log-optimality of e-processes and sequential tests</i>	Stanford, CA 2025
MBZUAI-Berkeley Workshop <i>Anytime-valid off-policy inference for contextual bandits</i>	Abu Dhabi, UAE 2025
Inria/Sierra Seminar <i>Anytime-valid inference and uniform central limit theory</i>	Paris, France 2025
International Seminar on Selective Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Virtual 2024
CLIMB Workshop <i>Election audits via anytime-valid inference</i>	Berkeley, CA 2024
Workshop on Game-Theoretic Statistical Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Oberwolfach, Germany 2024
Fienberg Student Research Workshop at Carnegie Mellon University <i>Election audits via anytime-valid inference</i>	Pittsburgh, PA 2024
Joint Statistical Meetings <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
International Conference on Machine Learning <i>Extensions of randomized response for private confidence sets</i>	Honolulu, HI 2023
Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen) <i>Anytime-valid off-policy inference for contextual bandits</i>	Amsterdam, Netherlands 2023
University of Toronto / The Vector Institute (host: Dan Roy) <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
Copenhagen Causality Lab, University of Copenhagen <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Virtual 2023
Conference on Digital Experimentation (CODE@MIT) <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Cambridge, MA 2022
Microsoft Research Reinforcement Learning Discussion Group <i>Anytime-valid contextual bandit inference</i>	Virtual 2022
Computing and Mathematical Sciences, California Institute of Technology (host: Houman Owhadi) <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Microsoft Research <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Safe, Anytime-Valid Inference (SAVI) Workshop <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Eindhoven, Netherlands 2022
Carnegie Mellon University Computer Science Theory Lunch <i>Estimating means of bounded random variables by betting</i>	Pittsburgh, PA 2021

International Seminar on Distribution-Free Statistics <i>Estimating means of bounded random variables by betting</i>	Virtual 2021
E-Vote-ID: The International Conference for Electronic Voting <i>RiLACS: Risk-limiting audits via confidence sequences</i>	Virtual 2021
Spotify Experimentation Platform Team <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Vinted Science and Analytics Meetup <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Other talks	
International Conference on Statistics and Data Science (ICSIDS) <i>Concentration inequalities for strong laws of large numbers</i>	Seville, Spain 2025
Statistical Society of Canada meeting <i>Distribution-uniform strong laws of large numbers</i> Recipient of the Probability Section's Student Presentation Award	St. John's, NL 2024
International Conference on Statistics and Data Science (ICSIDS) <i>Distribution-uniform anytime-valid inference</i>	Lisbon, Portugal 2023
Waterloo Student Conference in Statistics, Actuarial Science, and Finance <i>Estimating means of bounded random variables by betting</i> Recipient of the Student Presentation Award	Waterloo, Ontario 2022
TPDP: Theory and Practice of Differential Privacy Workshop <i>Locally private nonparametric confidence intervals and sequences</i>	Baltimore, MD 2022
Statistical Society of Canada (SSC) Annual Meeting <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2022
NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Confidence sequences for sampling without replacement</i>	Virtual 2020
Statistical Society of Canada (SSC) Annual Meeting <i>Multi-state models for chronic kidney disease prevalence projections in Ontario</i>	St. Catherines, Ontario 2016